

The Work–Impulse Hinge

The same force produces different accumulated effects through space and through time

Working rule: Identify the force, then ask what it is accumulated against. Through a displacement it transfers energy; through a time interval it changes momentum.

Name: _____

Class: _____

Hinge worksheet series

1. Constant force: energy through displacement

A crate is pulled 6.0 m along a horizontal floor by a constant force of 85 N acting at 30° above the horizontal. Friction is 28 N.

A. Represent the forces and displacement

Draw a free-body diagram. Add the displacement vector and mark the angle between the pulling force and displacement.

B. Calculate work by each horizontal force

Calculate the work done by the pulling force and by friction. Then find the net work.

$$W = Fs \cos \theta.$$

C. Join work to energy

Use the work–energy theorem to find the change in kinetic energy. If the crate starts from rest and has mass 12 kg, calculate its final speed.

D. Interrogate the scalar product

Why does the vertical component of the pull do no work on the crate? What would change if the displacement were inclined?

2. Force–time graph: momentum through time

A 0.20 kg ball initially moves to the right at 8.0 m s^{−1}. During impact, the force on it acts to the left. The force rises linearly from zero to 600 N in 4.0 ms and then falls linearly to zero over the next 6.0 ms.

A. Represent the interaction

Sketch the force–time graph with a clear sign convention. Mark the total contact time and the area that represents impulse.

B. Calculate impulse

Find the impulse from the area under the graph. Include its sign and units.

$$J = \Delta p = \int F \, dt.$$

C. Join impulse to momentum

Calculate the final momentum and velocity of the ball. State whether it rebounds.

D. Interrogate force and duration

For the same impulse, how would increasing the stopping time affect the average force? Give one practical application.

3. One interaction viewed in space and time

A 900 kg car travelling at 20 m s^{−1} is brought to rest by an approximately constant braking force over 5.0 s.

A. Use the time view

Calculate the change in momentum, impulse and average braking force. State the direction of the force.

B. Use the space view

Calculate the change in kinetic energy and the work done by the brakes. Use the force to determine the stopping distance.

C. Compare the two accumulations

Explain why impulse is a vector while work is a scalar. Show how the same braking force appears in both calculations but produces different outputs.

D. State the modelling assumptions

Which assumptions allow one constant average force to represent the braking process? What evidence would challenge the model?

Synthesis: force is shared; the accumulation variable determines the result

	Quantity	Accumulated against	Mathematical operation	Resulting change
1. Complete the comparison.	Work			2. Develop the central claim. Respond to: “ A force does not have one accumulated effect: through space it changes energy; through time it changes momentum.” Use the terms <i>displacement</i> , <i>time interval</i> , <i>scalar product</i> , <i>vector</i> , <i>area under a graph</i> , <i>system boundary</i> , and <i>sign convention</i> . _____ _____ _____
	Impulse			

Teacher guide and indicative answers

The Work–Impulse Hinge

1. Work through displacement

Pulling force:

$$W_{\text{pull}} = 85(6.0) \cos 30^\circ = 4.42 \times 10^2 \text{ J}.$$

Friction:

$$W_f = -28(6.0) = -168 \text{ J}.$$

Hence

$$W_{\text{net}} = 274 \text{ J} = \Delta E_k.$$

Starting from rest,

$$\frac{1}{2}mv^2 = 274 \quad \Rightarrow \quad v = 6.76 \text{ m s}^{-1}.$$

The normal force and weight are perpendicular to the horizontal displacement, so each does zero work. The vertical component of the pull likewise has zero scalar product with the displacement. **Key distinction.** Work depends on force and displacement geometry. It is energy transferred across the chosen system boundary and is a scalar.

2. Impulse through time

The graph is triangular with base $10.0 \text{ ms} = 1.00 \times 10^{-2} \text{ s}$ and height 600 N . Taking right as positive,

$$J = -\frac{1}{2}(600)(1.00 \times 10^{-2}) = -3.0 \text{ N s}.$$

Initial momentum:

$$p_i = (0.20)(8.0) = 1.6 \text{ kg m s}^{-1}.$$

Thus

$$p_f = p_i + J = 1.6 - 3.0 = -1.4 \text{ kg m s}^{-1},$$

$$v_f = \frac{-1.4}{0.20} = -7.0 \text{ m s}^{-1}.$$

The negative velocity shows that the ball rebounds.

For fixed impulse,

$$F_{\text{av}} = \frac{\Delta p}{\Delta t}.$$

Increasing the interaction time reduces the average force. Examples include airbags, crumple zones, helmets and bending the knees when landing.

3. The same braking interaction

Initial momentum is

$$p_i = (900)(20) = 1.80 \times 10^4 \text{ kg m s}^{-1}.$$

Taking forward as positive,

$$\Delta p = -1.80 \times 10^4 \text{ kg m s}^{-1},$$

$$F_{\text{av}} = \frac{\Delta p}{\Delta t} = -3.60 \times 10^3 \text{ N}.$$

The kinetic-energy change is

$$\Delta E_k = 0 - \frac{1}{2}(900)(20)^2 = -1.80 \times 10^5 \text{ J}.$$

Hence the work done by the braking force is $-1.80 \times 10^5 \text{ J}$. With constant force,

$$W = Fs \quad \Rightarrow \quad s = \frac{1.80 \times 10^5}{3.60 \times 10^3} = 50 \text{ m}.$$

Impulse retains direction because momentum and force are vectors. Work is a scalar because it is formed from the scalar product $\mathbf{F} \cdot \Delta \mathbf{r}$.

The constant-force model assumes approximately uniform deceleration, level motion, fixed mass and negligible variation in braking or drag forces.

Teacher synthesis: two integrals of force, two conserved quantities

	Quantity	Accumulated against	Output	Character	
Indicative comparison.	Work	displacement	change in energy	scalar	Core interpretation. The force law supplies the interaction. The accumulation variable determines what the interaction changes. A force acting through displacement transfers energy; the same force acting through time changes momentum. The relations do not compete, and a single event can be analysed using both.
	Impulse	time	change in momentum	vector	
For variable force, each quantity is an area under a different graph:					
$W = \int F_{\parallel} ds, \qquad J = \int F dt.$					
Expected conclusion. Students should distinguish the physical quantity being accumulated, the geometry or sign convention, and the system boundary. Similar-looking products Fs and $F\Delta t$ encode different mathematical operations and different conservation structures.					